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Wild Salmon Policy 2018 to 2022 Implementation Plan

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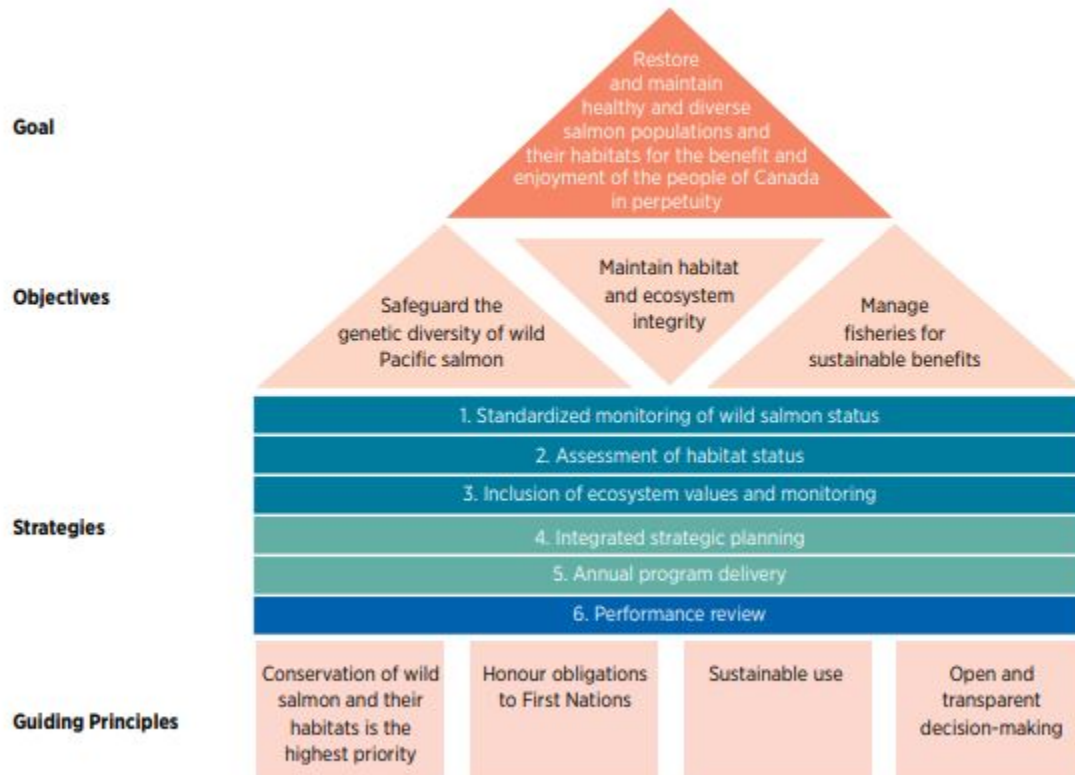
Annex 1: Acronyms

Acronym	Expansion
ADGT	Aquatic Diagnostics, Genomics and Technology
BCARP	BC Aquaculture Regulatory Program
CEAA	Canadian Environmental Assessment Act
CESD	Commissioner of Environment and Sustainable Development
CFIA	Canadian Food Inspection Agency
CIP	Community Involvement Program
CIRRP	Community Involvement and Resource Restoration Program
COSEWIC	Committee On the Status of Endangered Wildlife In Canada
CRF	Coastal Restoration Fund
CSAS	Canadian Scientific Advisory Secretariat
CU	Conservation Unit
DFO	Fisheries and Oceans Canada
DU	Designatable Unit (under SARA)
ENGO	Environmental Non-Governmental Organization
ESD	Ecosystem Science Division
FHMP	Fish Health Management Plan
FLNRO	Forest, Lands, Natural Resource Operations and Rural Development
FNFA	First Nations Final Agreement
FPP	Fisheries Protection Program
FRPA	Forest and Range Practices Act
FSC	Food, Social and Ceremonial
IFMP	Integrated Fisheries Management Plan
IKS	Indigenous Knowledge Systems

Acronym	Expansion
MAACFA	Minister of Agriculture's Advisory Council on Finfish Aquaculture
MU	Management Unit
NuSEDS	New Salmon Escapement Database Systems
NGO	Non-Governmental Organization
OPP	Oceans Protection Plan
OSD	Ocean Science Division
PD	Program Delivery
PIP	Public Involvement Program
PNCIMA	Pacific North Coast Integrated Management Area
PSC	Pacific Salmon Commission
PSE	Pacific Salmon Explorer
PSF	Pacific Salmon Foundation
PST	Pacific Salmon Treaty
QAMS	Quantitative Assessment Methods Section
RAMS	Risk Assessment Model for Salmon
RAR	Riparian Areas Regulation
REEFF	Regional Ecosystem Effects on Fish and Fisheries Section
RKW	Resident Killer Whale
RMPD	Resource Management Program Delivery
RRU	Resource Restoration Unit
SARA	Species At Risk Act
SEP	Salmonid Enhancement Program
SMCS	Salmon Management and Client Services

Acronym	Expansion
StAR	Stock Assessment And Research
WCVI	West Coast of Vancouver Island
WSA	Water Sustainability Act
WSP	Wild Salmon Policy
YESSA	Yukon Environmental and Socio-Economic Assessment Act
YFWMB	Yukon Fish and Wildlife Management Board
YRSA	Yukon River Salmon Agreement
YSSC	Yukon Salmon Sub-Committee

Annex 2: Goal and guiding principles of the WSP



Source: WSP, 2005

► Long text version

The 2005 Wild Salmon Policy house-shaped graphic (above) is intended to provide an overview of the WSP and illustrate that the Goal, Objectives, and Strategies of the Policy are built on, and supported by, a strong foundation of fundamental Guiding Principles.

While many of the aspects of the graphic and its intention remain true, with the passage of time some elements have, and will continue to evolve. In particular, the Guiding Principle “Honour obligations to First Nations” is outdated and should be considered to more broadly represent the Government of Canada’s commitment to the United Nations Declaration on the Rights of Indigenous Peoples, and a renewed relationship with Indigenous peoples in Canada based on the recognition of rights, respect, cooperation and partnership.

Annex 3: WSP implementation over time

Year	Implementation
1999	WSP consultations and development begin
2005	Wild Salmon Policy published after 5 years of consultations
2008	CSAS Science Advisory Report published: Framework for characterizing Conservation Units of Pacific salmon (<i>Oncorhynchus</i> spp.) for implementing the Wild Salmon Policy
2009	• Cohen Commission called: Inquiry into the Decline of Sockeye Salmon in the Fraser River • CSAS Science Advisory Report published: Framework for implementation of the Wild Salmon Policy: Initial lists of Conservation Units for British Columbia • Sustainable Fisheries Framework announced with a suite of management policies to ensure conservation, sustainable use and economic prosperity
2010	Pacific Aquaculture Regulations introduced as DFO assumes primary responsibility for aquaculture in BC
2011	Gardner Pinfold independent performance review of WSP implementation released

Year	Implementation
2012	• Cohen Commission releases report with 75 recommendations, including 8 directing DFO to act further on WSP • Changes to the Fisheries Act
2016–2017	WSP implementation consultations with First Nations, key partners and the public
2018	• Changes to the Fisheries Act • WSP 2018 to 2022 Implementation Plan published
2019	Annual Report on WSP implementation anticipated
2020	WSP Implementation Plan updates to include Fisheries Act program changes, as required
2022	5-year review of WSP 2018 to 2022 Implementation Plan to be undertaken
2023 and Beyond	Ongoing implementation of WSP

Annex 4: WSP implementation tools and resources

Policy and general information

- [Wild Salmon Policy \(2005\)](#)
- [Fisheries and Oceans Canada](#)

Data tools

- [NuSEDS](#)
DFO stores up-to-date information on the number and identity of CUs in the New Salmon Escapement Database System (NuSEDS), which holds data on adult salmon escapement. This centralized database is available to the public through the Government of Canada's Open Data Portal. As the numbers and boundaries of CUs change over time, the database will be updated.
- [Pacific Salmon Explorer](#)
The Pacific Salmon Explorer gives a high-level overview of salmon CUs, salmon abundance, status and trends over time, and cumulative pressures on freshwater and estuarine salmon habitats.

Published research

- [Canadian Science Advisory Secretariat \(CSAS\)](#)
This site houses CSAS publications dating back to 1977, and publishes the science advisory schedule.
- [Canadian Technical Report of Fisheries and Aquatic Sciences](#)
State of the physical, biological and selected fisheries resources of Pacific Canadian marine ecosystems in 2016.
- [Government of BC's Environmental Information Resources System for Biodiversity \(EIRS BDP\)](#)
This database contains a range of environmental and natural resource information, including publications on British Columbia's species and their habitats. Including:
 - [A Framework for Conducting Effectiveness Evaluations of Watershed Restoration Projects](#)
[Guidelines for Planning Watershed Restoration Projects](#)
 - [The Keogh and Waukwaas Rivers Paired Watershed Study for B.C.'s Watershed Restoration Program: Juvenile Salmonid Enumeration and Growth 1997](#)

Programs

- [Aquaculture](#)
Aquaculture science and research website.
 - [Aquaculture in British Columbia](#)
DFO's Aquaculture in British Columbia website outlines the regulations, environmental management requirements and compliance and monitoring activities that are in place to ensure that the industry operates in a sustainable manner.
 - [Information on heart and skeletal muscle inflammation \(HSMI\) and piscine reovirus \(PRV\)](#)
- [Fisheries](#)
- [Fisheries Protection Program \(FPP\)](#)
- [Recreational Fisheries Conservation Partnerships Program](#)
- [Salmonid Enhancement Program \(SEP\)](#)
 - [SEP Community Advisors](#)
 - [Resource Restoration Unit](#)
 - [Fish Habitat Restoration Initiative](#)
 - [Coastal Restoration Fund](#)
- [Species at Risk](#)
- [Oceans Program](#)
- [Science and Research](#)

Agreements and regulations impacting wild Pacific salmon

- [Fisheries Act Updates](#)

- Key BC Legislation
 - [BC Water Sustainability Act](#)
 - [BC Riparian Areas Regulation](#)
 - [BC Forest and Range Practices Act](#)
 - [BC Oil and Gas Activities Act](#)
- BC Treaty Commission Website and Final Agreements
 - [BC Treaty Commission Website](#)
 - [Nisga'a Final Agreement](#)
 - [Tsawwassen Final Agreement](#)
 - [Maa-'nulth Final Agreement](#)
 - [Tla'amin Final Agreement](#)

Key Yukon Legislation Impacting Wild Salmon

- [Yukon Environment Act](#)
- [Yukon Waters Act](#)
- [Yukon Forest Resources Act](#)

Yukon First Nations

- [Yukon First Nations Final Agreements](#)

Species at Risk Act (SARA)

- [Public Registry](#)

Steelhead

- [Provincial Framework for Steelhead Management in British Columbia](#)
- [DFO Species Information](#)

Cohen Commission

- [The Cohen Response](#)
Includes a summary of actions taken on the Cohen Commission recommendations.
- [Cohen Commission Report](#)
The Final Report of the Cohen Commission of Inquiry into the Decline of Sockeye Salmon on the Fraser River released in 2012

Annex 5: Glossary

Term	Definition
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Term	Definition
Anadromous fish	Species that migrate from marine to freshwater environments to spawn.
Anthropogenic	of, relating to, or resulting from the influence of human beings on nature. Aquaculture: The farming of aquatic organisms in the marine environment or freshwater.
Benchmark	A standard (quantified metric) against which habitat or population status can be measured or judged and by which status can be compared over time and space to determine the risk of adverse effects. For instance, biological benchmarks demarcate zones of population status based on conservation and production considerations (Holt and Irvine 2013).
Biodiversity or biological diversity	The full range of variety and variability within and among living organisms and the ecological complexes in which they occur; encompasses diversity at the ecosystem, community, species, and genetic levels and the interaction of these components.
Biogeographic distribution	The distribution of species and ecosystems in geographic space and through geological time.
Biosecurity	procedures intended to protect humans, animals or the larger environment against disease or harmful biological agents.
Bycatch	The unwanted, non-target or discarded fish and other marine creatures caught during commercial fishing activities. Closed Containment Aquaculture System: Any system of fish production that creates a controlled interface between the cultured animal and the external natural environment (eg. land based aquaculture farms).
Conservation	The protection, maintenance, and rehabilitation of genetic diversity, species, and ecosystems to sustain biodiversity and the continuance of evolutionary and natural production processes.
Conservation Unit (CU)	A group of wild salmon sufficiently isolated from other groups that, if extirpated, is very unlikely to re-establish naturally within an acceptable timeframe (eg. a human lifetime or a specified number of salmon generations).

Term	Definition
COSEWIC	Committee on the Status of Endangered Wildlife in Canada; an independent advisory panel to the Minister of Environment and Climate Change Canada that meets twice a year to assess the status of wildlife species at risk of extinction. Members are wildlife biology experts from academia, government, non-governmental organizations and the private sector responsible for designating wildlife species in danger of disappearing from Canada.
Cumulative effects	Changes to environmental, social and economic values caused by the combined effect of past to present activities and events. Within a WSP context, status changes to wild salmon conservation units and the ecosystem units required to sustain them are key cumulative effects indicators.
Ecosystem	A community of organisms plus physical and chemical conditions of their environment interacting as an ecological unit. Ecosystem-based approach: A strategy for the integrated management of land, water and living resources that promotes conservation and sustainable use in an equitable way; Conservation of ecosystem structures, processes and interactions through sustainable use.
Ecosystem-based approach	A strategy for the integrated management of land, water and living resources that promotes conservation and sustainable use in an equitable way; conservation of ecosystem structures, processes and interactions through sustainable use.
Ecosystem integrity	The physical, chemical and biological structure and processes that characterize specific ecosystems.
Ecosystem values	Attributes or properties of ecosystems that society wishes to sustain. Freshwater ecosystem integrity, salmon biodiversity and sustainable fisheries for salmon are examples of ecosystem values within a WSP context.
Enhancement	The application of biological and technical knowledge and capabilities to increase the productivity of fish stocks. It may be achieved by altering habitat attributes (e.g. habitat restoration) or by using fish culture techniques (e.g. hatcheries, spawning channels). In the context of the WSP, only salmon originating directly from hatcheries and managed spawning channels will be considered enhanced.

Term	Definition
Epigenetic	Heritable changes in gene function that do not involve changes in the DNA sequence.
Escapement	The number of mature salmon that pass through (or escape) fisheries and return to fresh water to spawn. Extirpation: The local extinction of a species. Flora and Fauna: Plant life and animal life, respectively.
Fecundity	The number of eggs an animal produces during each reproductive cycle; the potential reproductive capacity of an organism or population (similar to fertility).
Fish habitat	Spawning grounds and any other areas, including nursery, rearing, food supply and migration areas, on which fish depend directly or indirectly in order to carry out their life processes.
Fishery Reference Point (FRP)	A point at which management actions occur. An example is the abundance of returning adults above which targeted harvest is considered. A limit reference point and an upper reference point define the boundaries for managing to protect stock status. FRPs are distinct from biological benchmarks.
Fry	Salmon that have emerged from gravel, completed yolk absorption, remained in freshwater streams, and are less than a few months old.
Genetic diversity	The variation at the level of individual genes, and provides a mechanism for populations to adapt to their ever-changing environment. It refers to the differences in genetic make-up among distinct species and to genetic variations within a single species.
Genetic traits	Characteristics or attributes of an organism that are expressed by genes and/or influenced by the environment. Traits include physical attributes and behavioural characteristics.
Habitat restoration	Physical or chemical manipulations to create a natural or normative reference state of habitat that has been altered, disrupted, or degraded. Successful restoration increases the capability of previously damaged habitat to sustain fish production.
Indicator species	A species whose presence, absence or abundance, relates to a specific environmental condition.

Term	Definition
Integrated Fisheries Management Plans (IFMPs)	Used by DFO to guide the conservation and sustainable use of marine resources. An IFMP is developed to manage the fishery of a particular species in a given region. IFMPs combine the best available science on a species with industry data on capacity and methods for harvesting that species.
Invasive species	A plant, fungus or animal species that is not native to a specific location and has negative effects on our economy, environment and/or health.
Keystone species	A species that has a disproportionately large effect on its environment relative to its abundance. Such species are described as playing a critical role in maintaining the structure of an ecological community, affecting many other organisms in an ecosystem and helping to determine the types and numbers of various other species in the community. Pacific salmon are regarded as keystone species due to their disproportionate impact as sources of food and nutrients that demonstrably limit the distribution and abundance of a wide range of both plant and animal species in freshwater and associated terrestrial ecosystems.
Lower benchmark	A reference point in biological status associated with significant losses in production between the Amber and Red zones, and which allows for a substantial buffer between it and any level of abundance that could lead to a CU being considered at risk of extinction.
Metric	A quantifiable measure.
Multi-trophic	A trophic level is a functional classification of organisms in a community according to feeding relationships. Ecosystems contain multiple trophic levels consisting of plants, grazers, predators, scavengers etc.
Natal stream	Specific locality of birth of a fish or other aquatic animal.
Pathogen	A bacterium, virus or other microorganisms that can cause disease.
Pelagic	Related to, living, or occurring in the open-ocean.
Population	A group of interbreeding organisms that is relatively isolated (i.e. demographically uncoupled) from other such groups and is likely adapted to the local habitat.

Term	Definition
Precautionary approach	When used in an advisory context in support of decisionmaking by the Government of Canada, this term conveys the sense that the advice is provided in situations of high scientific uncertainty. It is intended to promote actions that would result in a low probability of harm that is serious or difficult to reverse.
Predator-scavenger complex (PSC)	An ecological community consisting of many species of animals that obtain a significant portion of their seasonal to annual food requirements by killing or scavenging and then eating a common species. In British Columbia the salmon PSC comprises at least 23 species of mammals and birds that consume adult fish returning to British Columbia rivers and streams.
Productivity	A measure of the amount of energy (or material) formed by an individual population or community in a specific amount of time. Within a WSP context, some common measures of productivity include the number of fry, smolts or subsequent adults (i.e. recruits) produced on average per spawning adult of the previous generation.
Resource management	Departmental actions, policies and programs affecting wild Pacific salmon directly or indirectly through their habitats and ecosystems.
Returns	Total number of adults returning to freshwater to spawn in a given year, including those caught in fisheries.
Riparian	Pertaining to anything connected to with or adjacent to the bank of the stream or other body of water.
Salmonid	A group of fish that includes salmon, trout, and char, belonging to the taxonomic Family Salmonidae.
Spawner	A reproductively mature individual fish returning to freshwater to spawn.
Species	The fundamental category of taxonomic classification consisting of organisms grouped by virtue of their common attributes and capable of interbreeding. A taxonomic species is equivalent to the term “species” but the phrase may be used to indicate the collective species throughout its distribution.
Stewardship	Acting responsibly to conserve fish and their habitat for present and future generations.

Term	Definition
Stock assessment	The use of various statistical and mathematical calculations to make quantitative predictions about the reactions of fish populations to alternative management choices.
Terminal fisheries	Fisheries that occur near or in fresh water (e.g. at a river's mouth) where the targeted species or stock has returned to spawn.
Wild salmon	Salmon are considered "wild" if they have spent their entire life cycle in the wild and originate from parents that were also produced by natural spawning and continuously lived in the wild.

Annex 6: References

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